

Canonical Class-II K current and the first renormalisation split

TECT verification-first repository · claim A6-CLASSII-K-COMPOSITE-DEFINITION

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1. Purpose and scope

The previous ultraviolet calculation identified a positive linear contraction in the derivative Class-II energy, but it deliberately left two logically different questions open. First, does the current K_A itself have a canonical continuum definition? Second, can the leading energy contraction be subtracted while preserving stability, or can an unsubtracted measure concentrate on its zero set?

This note closes the first question at fixed positive density floor and inside a declared symmetry-preserving spectral regulator class. It then reviews the other questions separately. The literal leading- W_ε subtraction is ruled out as a cutoff-uniform fixed-parameter coercivity route. Two local bare-concentration proxies are solved exactly, but no conclusion is drawn for the spatially correlated Gibbs field.

The domain is the three-torus of side $L = 16$. The covariance and all production parameters are imported from the content-addressed A1 production manifest. The density floor is fixed at $\varepsilon_\rho = 10^{-12}$. Floor removal, infinite volume, a phase transition, and BCC existence or selection are outside scope.

2. Statement

Realify the three-component complex field as $X : \mathbb{T}_L^3 \rightarrow \mathbb{R}^6$. Let S_A be the real symmetric representation of an embedded Pauli generator and set

$$\rho(X) = X^\top X, \quad m_A(X) = X^\top S_A X, \quad q_A(X) = \frac{m_A(X)}{\rho(X) + \varepsilon_\rho}.$$

The relevant one-form is

$$\omega_A = dm_A - q_A, d\rho = (\rho + \varepsilon_\rho)dq_A, \quad \omega_{A,a}(X) = 2\{(S_A X)_a - q_A(X)X_a\}.$$

Theorem 1 (Fixed-floor canonical spectral current). *Suppose the centered stationary six-real Gaussian field has an even symmetric Fourier covariance $Q(k)$ satisfying*

$$c\langle k \rangle^{-4}I_6 \leq Q(k) \leq C\langle k \rangle^{-4}I_6.$$

Let R_Λ be a scalar Fourier multiplier common to all six coordinates, real and even, uniformly bounded, converging pointwise to one, and having compact spectral support or a uniform Schwartz tail. Let $X_\Lambda = R_\Lambda X$ and evaluate nonlinear products by exact convolution or sufficient de-aliasing. For every $p < \infty$ and $\kappa > 0$,

$$K_{A,i}^{(\Lambda)} = \omega_A(X_\Lambda) \partial_i X_\Lambda \longrightarrow K_{A,i} \quad \text{in } L^p(\Omega; \mathcal{C}^{-1/2-\kappa}).$$

The limit is independent of the multiplier inside this class and is the canonical geometric, equivalently symmetric-Stratonovich, reconstruction of the one-form current. Along a dyadic subsequence the convergence is almost sure. No divergent K_A -level subtraction occurs.

The limit identities are written

$$K_A = \nabla m_A - q_A \diamond \nabla \rho = (\rho + \varepsilon_\rho) \diamond \nabla q_A,$$

where $\diamond$ denotes the canonical geometric product. It is not an ordinary product of arbitrary distributions. The theorem defines K_A but does not define $J_A K_A$, $|K_A|^2$, or the Class-II energy.

3. Setting and proof of the composite theorem

3.1 Gaussian regularity

The production symbol has leading part $Y|k|^4 I_6$. On a dyadic shell,

$$\mathbb{E}|\Delta_j X(x)|^2 \lesssim \sum_{|k| \asymp 2^j} \langle k \rangle^{-4} \lesssim 2^{-j}.$$

Gaussian moment equivalence, Bernstein estimates, and the usual dyadic summation imply

$$X \in \bigcap_{\eta > 0} \mathcal{C}^{1/2-\eta} \quad \text{almost surely and in every finite } L^p(\Omega),$$

while $\partial_i X \in \mathcal{C}^{-1/2-\eta}$. The endpoint $H^{1/2}$ is not claimed.

3.2 The canonical second level

Fix $\alpha \in (1/3, 1/2)$. The only nonclassical input required by the one-form current is the resonant product

$$\mathbb{X}_i^{ab,\Lambda} = X_\Lambda^a \circ \partial_i X_\Lambda^b.$$

The resonant restriction has $|k| \asymp |\ell|$. Wick's identity and the covariance bound give, uniformly in the admitted multipliers,

$$\begin{aligned} \mathbb{E} \left| \widehat{\mathbb{X}_i^{ab,\Lambda}}(p) \right|^2 &\lesssim \sum_{\substack{k+\ell=p \\ |k| \asymp |\ell|}} \langle k \rangle^{-4} \ell_i^2 \langle \ell \rangle^{-4} \\ &\lesssim \langle p \rangle^{-3}. \end{aligned}$$

The last estimate is obtained by splitting $|k| \lesssim |p|$ from $|k| > |p|$. The first part has $O(|p|^3)$ summands of size $O(|p|^{-6})$; the second is bounded by $\sum_{|k| > |p|} O(|k|^{-6}) = O(|p|^{-3})$.

At fixed output mode the cutoff tail obeys

$$\mathbb{E} \left| \widehat{\mathbb{X}_i^{ab,\Lambda}} - \mathbb{X}_i^{ab}(p) \right|^2 \lesssim \sum_{|k| > c\Lambda} |k|^{-6} = O(\Lambda^{-3}).$$

Dominated convergence handles two different admitted multipliers coupled to the same Gaussian modes. Littlewood-Paley summation and hypercontractivity of the second chaos yield convergence in $L^p(\Omega; \mathcal{C}^{-\eta})$ for every $p < \infty$ and $\eta > 0$. Summability of dyadic cutoff increments then gives almost-sure convergence along dyadic subsequences.

For every admitted regulator,

$$\mathbb{E}[X_\Lambda^a \partial_i X_\Lambda^b] = \partial_i C_\Lambda^{ab}(0) = 0,$$

because the covariance and common multiplier are even. Thus the second level has no local divergent contraction to subtract. Its symmetric part is fixed by

$$X^a \diamond \partial_i X^b + X^b \diamond \partial_i X^a = \partial_i (X^a X^b).$$

3.3 Nonlinear reconstruction

The positive floor makes every derivative of ω_A smooth with at most polynomial growth controlled by powers of ε_ρ^{-1} . First order parilinearisation gives

$$\omega_{A,a}(X) = T_{\partial_b \omega_{A,a}(X)} X^b + R_{A,a}, \quad R_{A,a} \in \mathcal{C}^{2\alpha}.$$

All paraproducts with $\partial_i X^a$ are classical except the resonant term, which is reconstructed from

$$\partial_b \omega_{A,a}(X) \mathbb{X}_i^{ba}.$$

The coefficient is in $\mathcal{C}^\alpha$ and the second level is in $\mathcal{C}^{-\eta}$, so this product is classical for $\eta < \alpha$. Moreover,

$$R_{A,a} \circ \partial_i X^a \in \mathcal{C}^{3\alpha-1},$$

which is well defined because $3\alpha - 1 > 0$. The reconstruction is a continuous map of the enhanced Gaussian field $(X, \mathbb{X})$. Convergence and scheme independence of the enhancement therefore imply the asserted convergence of K_A . The least regular low-high paraproduct has regularity $\alpha - 1$, giving the declared $\mathcal{C}^{-1/2-\kappa}$ target.

This is the same second-level mechanism used in paracontrolled calculus; see Gubinelli, Imkeller, and Perkowski, <https://arxiv.org/abs/1210.2684>. Canonical Gaussian rough lifts and approximation convergence are developed by Friz and Victoir, <https://arxiv.org/abs/0711.0668>. The equality of Gaussian rough and symmetric-Stratonovich integrals under suitable regularity is treated by Ohashi and Russo, <https://arxiv.org/abs/2206.06865>; correction formulas relative to Skorohod integration are discussed by Cass and Lim, <https://arxiv.org/abs/1604.06846>. The Fourier estimates above are the specialised argument used for the pinned torus field.

3.4 Why the enhancement is load-bearing

Direct differentiation gives

$$(d\omega_A)_{ab} = \frac{4}{\rho + \varepsilon_\rho} \{X_a(S_A X)_b - X_b(S_A X)_a\}.$$

For the real two-coordinate slice of $S_3 = \text{diag}(1, -1)$,

$$\rho = x^2 + y^2, \quad m_3 = x^2 - y^2,$$

and

$$d\omega_3 = -\frac{8xy}{x^2 + y^2 + \varepsilon_\rho} dx \wedge dy.$$

This is generically nonzero. Two enhancements differing by antisymmetric area Γ_i^{ab} can change the current by

$$\Delta K_{A,i} = \frac{1}{2} \sum_{a,b} (d\omega_A)_{ab}(X) \Gamma_i^{ab}.$$

The common-even spectral class fixes $\Gamma = 0$ relative to the canonical Gaussian lift. A component-split asymmetric approximation is outside this class and can have a finite anomaly. Therefore the theorem is scheme controlled, not an unrestricted universality claim.

4. Counterterm and bare-concentration split

4.1 Exact W_ε bounds

Write

$$Q_{II} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad s = |\Psi_1|^2 + |\Psi_2|^2,$$

and let $v_A = (T_A - q_A I)\Psi$. The leading contraction is

$$W_\varepsilon = 3 \sum_A \begin{pmatrix} T_A \Psi \\ v_A \end{pmatrix}^* Q_{II} \begin{pmatrix} T_A \Psi \\ v_A \end{pmatrix}.$$

Since $Q_{II} > 0$ and $\sum_A |T_A \Psi|^2 = 3s$,

$$W_\varepsilon \geq 9\lambda_{\min}(Q_{II})s.$$

The exact Fierz form is

$$W_\varepsilon = 9(a + 2b + c)s - \frac{6bs^2}{\rho + \varepsilon_\rho} - \frac{3cs^2(\rho + 2\varepsilon_\rho)}{(\rho + \varepsilon_\rho)^2},$$

so, because $b, c > 0$,

$$W_\varepsilon \leq 9(a + 2b + c)s.$$

It follows that $W_\varepsilon = 0$ exactly when $\Psi_1 = \Psi_2 = 0$. This is an algebraic statement about the conditional leading contraction, not a pathwise lower bound on the full derivative energy.

4.2 No-go for literal fixed-parameter subtraction

Let $t_N = \delta_{\text{cube}} N$ and consider

$$F_N^{\text{naive}} = F_{\text{core}} + F_{\text{II},N} - t_N \int W_\varepsilon(\Psi) dx.$$

For a homogeneous field in the first component, every derivative current vanishes. Put $\rho = |\Psi_1|^2$. Then

$$W_\varepsilon(\sqrt{\rho}, e_1) = w_\infty \rho + O(\varepsilon_\rho), \quad w_\infty = 9a + 12b + 6c > 0,$$

while

$$\frac{F_{\text{core}}}{|\mathbb{T}^3|} = \frac{\gamma}{6}\rho^3 + \frac{\lambda}{4}\rho^2 + O(\rho).$$

Choose

$$\rho_N = \sqrt{\frac{2t_N w_\infty}{\gamma}}.$$

Substitution gives

$$\frac{F_N^{\text{naive}}(\sqrt{\rho_N} e_1)}{|\mathbb{T}^3|} = -\frac{2\sqrt{2}}{3\sqrt{\gamma}} w_\infty^{3/2} t_N^{3/2} + O(t_N),$$

hence there is no cutoff-uniform raw lower bound. In N variables,

$$\frac{|\Psi_N|}{N^{1/4}} \longrightarrow \left(\frac{2\delta_{\text{cube}} w_\infty}{\gamma} \right)^{1/4},$$

and the energy density divided by $N^{3/2}$ tends to a strictly negative constant. A vacuum shift changes the displayed minimum but not the normalised measure or the escaping amplitude. Sextic coercivity proves finite-cutoff normalisability only.

Adding the running relevant term $t_N h \int s$ with $h = 9(a + 2b + c)$ makes $t_N(hs - W_\varepsilon) \geq 0$. This is a possible stability repair, but it introduces a new renormalisation condition and favours collapse toward $s = 0$. It is not counterterm closure.

4.3 Mean-contraction local proxy

Let $z = (\Psi_1, \Psi_2) \in \mathbb{C}^2$, let $w = \Psi_3$, and let ν be a nondegenerate local Gaussian value law. Define

$$d\mu_t^W = Z_t^{-1} e^{-tW_\varepsilon(z,w)} d\nu(z,w).$$

For $z = y/\sqrt{t}$, fixed positive floor gives

$$tW_\varepsilon(y/\sqrt{t}, w) \longrightarrow h|y|^2, \quad h = 9(a + 2b + c).$$

The lower bound $W_\varepsilon \geq 9\lambda_{\min}s$ supplies domination. Laplace scaling in four real dimensions yields

$$Z_t \sim t^{-2} \frac{\pi^2 p_z(0)}{h^2},$$

and

$$(\sqrt{t}z, w) \implies (Y, W_0), \quad \frac{d\Pr_Y}{dy} = \left(\frac{h}{\pi}\right)^2 e^{-h|y|^2}.$$

Thus

$$t|z|^2 \implies \text{Gamma}(2, \text{rate} = h), \quad \mathbb{E}[t|z|^2] \longrightarrow \frac{2}{h}.$$

4.4 Exact derivative-integrated local proxy

The preceding proxy exponentiates the conditional mean and is not exact gradient integration. Instead let each of the three local complex gradient vectors have covariance tI_3 and define

$$G_t(\Psi) = \mathbb{E}_{\nabla\Psi} \exp\{-e_{II}(\Psi, \nabla\Psi)\}.$$

At $z = y/\sqrt{t}$, the floor makes $K_A - J_A \rightarrow 0$. The embedded Pauli identity gives, for one spatial direction,

$$\text{Cov}(J_A, J_B) = 2|y|^2 \delta_{AB}.$$

With $g = a + 2b + c$, a Gaussian determinant in three current coordinates and three spatial directions gives

$$G_t(y/\sqrt{t}, w) \longrightarrow G_\infty(y) = (1 + 2g|y|^2)^{-9/2}.$$

The lower eigenvalue of Q_{II} gives an integrable determinant bound. Hence the rescaled radial variable has density

$$f_R(r) = 35g^2 r (1 + 2gr)^{-9/2}, \quad r \geq 0,$$

with

$$\mathbb{E}R = \frac{2}{3g}, \quad \text{Var } R = \frac{14}{9g^2}.$$

This proxy also concentrates locally on $z = 0$, but with a different limiting law and a mean three times the mean-contraction proxy after substituting $h = 9g$. Neither local statement controls spatial entropy, mode coupling, partition functions, or tightness of the third component.

5. Numerical verification

The production coefficients are derived from the hash-pinned A1 manifest:

$$a = 0.004499999999998875, \quad b = 0.0018749999999995311, \quad c = 0.002343749999999414.$$

The principal derived values are

$$\begin{aligned}
9\lambda_{\min}(Q_{II}) &= 0.0113311035083346, \\
w_{\infty} &= 0.0770624999999807, \\
\frac{|\Psi_N|}{N^{1/4}} &\longrightarrow 0.198135127774404, \\
\frac{\inf F_N^{\text{naive}}/|\mathbb{T}^3|}{N^{3/2}} &\longrightarrow -3.26710156480221 \times 10^{-5}, \\
\mathbb{E}_{\text{mean}}[N|z|^2] &\longrightarrow 1294.94082886623, \\
\mathbb{E}_{\text{derivative}}[N|z|^2] &\longrightarrow 3884.82248659868.
\end{aligned}$$

The primary script uses FFT multiplicity convolution, exact Pauli/Fierz identities, finite-difference curvature, three even regulator families, the homogeneous trial, and direct determinant algebra. It passes 29/29. The non-importing audit uses direct (n_x, n_y, n_z) enumeration, a separately coded coefficient reduction, independent Gamma and beta-prime quadrature, and a synthetic positive q^{-4} covariance with a component-split phase regulator as a negative control. It passes 16/16. The one-command audit passes 64/64 in aggregate.

The primary fitted variance-tail slopes are

$$-2.98928 \quad (\text{cube}), \quad -3.01229 \quad (\text{ball}), \quad -3.01392 \quad (\text{smooth even}),$$

and the independent slopes are -3.06772 , -3.06468 , and -3.04209 . These values check the analytic N^{-3} variance tail; they do not replace the all-cutoff proof. The asymmetric negative control has a stable nonzero area drift and is not admitted by the theorem.

Reproduce with

```
python codes/foundations/a6_classii_k_composite_verify.py
```

Expected terminal lines are primary 29/29, independent 16/16, aggregate 64/64, and ‘A6-CLASSII-K-COMPOSITE-INTEGRATED-PASS’.

The previous A6 source was also re-run after correcting one documentation defect. Its function already returns $W = 3\sum_A(\cdots)$, so the leading conditional term is $\delta_{\text{cube}}NW$, not $3\delta_{\text{cube}}NW$. The correction changes no arithmetic; the original A6 integrated audit remains 44/44 PASS.

6. Devil’s-advocate

1. **VALID-with-mitigation:** deterministic Holder counting alone does not prove that the product diverges or is ambiguous. The closure instead uses the special Fourier-Wick second level and the one-form reconstruction.
2. **DISMISSED:** the rational denominator creates a hidden K -level subtraction. At fixed positive floor the one-form is smooth, and the only local second-level contraction vanishes by evenness.
3. **VALID-with-mitigation:** arbitrary regulator independence is false. The theorem is restricted to common real-even scalar multipliers. A component-split asymmetric negative control exhibits an anomaly.
4. **DISMISSED:** sharp spectral cutoffs invalidate the limit. The proof uses bounded pointwise-convergent multipliers and dominated Fourier-Wick sums, so smoothness of the multiplier is not required when convolution is exact.

5. **VALID-with-mitigation:** collocation aliasing can imitate scheme dependence. The scope requires exact convolution or sufficient de-aliasing; an aliased product is a falsifier, not an admitted implementation.
6. **DISMISSED:** defining K_A defines $|K_A|^2$. Squaring a $\mathcal{C}^{-1/2-}$ distribution is a separate renormalisation problem and is explicitly excluded.
7. **UPHELD:** literal subtraction of $\delta_{\text{cube}}NW_\varepsilon$ with fixed lower-order coefficients is not uniformly coercive. The exact homogeneous trial has an escaping $N^{1/4}$ amplitude and negative $N^{3/2}$ energy.
8. **VALID-with-mitigation:** $W_\varepsilon^{-1}(0)$ does not prove full Gibbs concentration. Only two local proxies are closed; the full-field problem is a separate gate.
9. **VALID-with-mitigation:** the constants are asymptotic and the couplings are weak, so existing $N = 32, 64, 128$ simulations need not display the local proxy limits. No practical crossover claim is made.
10. **VALID-with-mitigation:** no estimate is uniform as $\varepsilon_\rho \downarrow 0$. The floor is pinned at a positive value and removal is excluded.

7. Result footer

Result ID:	A6-CLASSII-K-COMPOSITE-DEFINITION
Precise statement:	Common real-even spectral approximations define one fixed-floor canonical geometric K_A current; literal fixed-parameter W subtraction is nonuniform, and two local bare proxies have distinct exact limits.
Scope:	$L=16$ production q^{-4} Gaussian; three complex fields; six-real convention; $\text{eps_rho}=1\text{e-}12$; common real-even scalar multipliers; exact/dealiased convolution.
Dependencies:	A1-PRODUCTION-FUNCTIONAL-REALISATION; A6-CLASSII-UV-POWER-COUNTING.
Evidence grade:	ANALYTIC, EXACT, EXECUTED, CONDITIONAL.
Reproduction command:	python codes/foundations/ a6_classii_k_composite_verify.py
Expected output:	primary 29/29; independent 16/16; aggregate 64/64; integrated PASS; exit 0.
Falsification gate:	failed Wick tail, scheme comparison, curvature, $W/\text{no-go}/\text{proxy identity}$, hash audit, or subprocess.
Tier before / after:	new / T5 PINNED-CLOSURE@FIXED-FLOOR-CANONICAL- SPECTRAL-K-LIFT.
No-overclaim statement:	No K -square, counterterm closure, full-field bare theorem, Gibbs measure, floor removal, arbitrary regulator universality, BCC, T6, or T7.
Next required action:	Design running relevant counterterms and separate full-field partition/tightness estimates.